

Name _____

Teacher _____



GOSFORD HIGH SCHOOL

2016

HIGHER SCHOOL CERTIFICATE

ASSESSMENT TASK 3

MATHEMATICS – EXTENSION 2

Duration- 60 minutes plus 5 minutes reading time

Integration Section	Only write on one side of the paper for your solutions. Staple this section together at the end of the exam.	/26
Conics Section	Start a new page for this question. – Only write on one side of the paper. Staple this section together at the end of the exam	/18
TOTAL		/44

Integration Section

Marks

1) Find

$$\int \frac{e^x}{1+e^{2x}} dx \text{ using the substitution } u = e^x$$

$$\frac{1}{2} \ln e^x$$

2

2) Evaluate

$$\int_0^{\frac{\pi}{2}} x \cos x \, dx$$

$$\frac{\pi}{2} - 1$$

2

3) Evaluate

$$\int_3^4 \frac{x^2 + x - 4}{x - 2} dx$$

$$\frac{13}{2} + \ln 2$$

3

4) (a) Find real numbers A, B and C such that

$$\frac{3x+7}{(x+1)(x+2)(x+3)} \equiv \frac{A}{x+1} + \frac{B}{x+2} + \frac{C}{x+3}$$

$$a=2$$

$$B=-1$$

$$C=-1$$

3

(b) Hence evaluate

$$\int_0^1 \frac{3x+7}{(x+1)(x+2)(x+3)} dx$$

$$\ln 2$$

3

5) By using the substitution $t = \tan \frac{x}{2}$

$$\text{evaluate } \int_0^{\frac{\pi}{3}} \frac{dx}{1 + \cos x - \sin x}$$

4

6) (a) Simplify $\sin(a + b) + \sin(a - b)$ 1

(b) Hence evaluate

$$\int_0^{\frac{\pi}{4}} \sin 5x \cos 3x \, dx$$
2

7)

Given $I_n = \int_0^1 x^n e^{-x} \, dx$

(a) Show that $I_n = -\frac{1}{e} + nI_{n-1}$ 3

(b) Hence find the exact value of

$$I_n = \int_0^1 x^3 e^{-x} \, dx$$
3

End of Integration section.

Go on to Conics section

Start a new page for the Conics section

Conics Section

Marks

- 1) Find the equation of the locus of a point $P(x,y)$ such that the distance from the point $S(1,0)$ is half its distance from the line $x=4$. 2

- 2) Find the coordinates of the foci and the equation of the directrices of the rectangular hyperbola $xy = 12$ 2

- 3) (a) Show that if $y = px + q$ is a tangent to the hyperbola $\frac{x^2}{a^2} - \frac{y^2}{b^2} = 1$ then $p^2a^2 - b^2 = q^2$ 3

- (b) Hence find the equations of the tangents from the point $(1,3)$ to the hyperbola $\frac{x^2}{4} - \frac{y^2}{15} = 1$ 3

- 4) An ellipse E can be described as the locus of a point moving such that the **sum** of its distances from two fixed points (**foci**) is constant.
Hint- Draw a diagram

- (a) If the two fixed points are $S(4, 0)$ and $S'(-4, 0)$ and the sum of the distance of $P(x, y)$ from these points is **10 units**, show that the equation of E is given by $\frac{x^2}{25} + \frac{y^2}{9} = 1$ 2

- (b) Verify that $x = 5\cos\theta$ and $y = 3\sin\theta$ are the parametric equations of E . 1

- (c) Find the equation of the normal to E at the point where $\theta = \frac{\pi}{6}$ 3

- (d) Determine the eccentricity of E and hence the equations of the directrices. 2

End of Examination

Integration Solutions

1) $\int \frac{e^x}{1+(e^x)^2} dx$

Let $u = e^x$

$$\frac{du}{dx} = e^x$$

$$\frac{du}{e^x} = dx \quad (1)$$

$$\int \frac{1}{1+u^2} du$$

$$= \tan^{-1} u + C$$

$$= \tan^{-1}(e^x) + C$$

(1)

2) Let $u = x$

$$\frac{dv}{dx} = \cos x$$

$$\frac{du}{dx} = 1$$

$$v = \sin x$$

$$\int_0^{\frac{\pi}{2}} x \cos x dx = \left[x \sin x \right]_0^{\frac{\pi}{2}} - \int_0^{\frac{\pi}{2}} \sin x dx \quad (1)$$

$$= \left[\frac{\pi}{2} \times 1 - 0 \right] + \left[\cos x \right]_0^{\frac{\pi}{2}}$$

$$= \frac{\pi}{2} + [0 - 1]$$

$$= \frac{\pi}{2} - 1 \quad (1)$$

$$\begin{array}{r}
 x+3 \\
 3/ \quad x-2 \overline{) x^2+x-4} \\
 \underline{x^2-2x} \\
 3x-4 \\
 \underline{3x-6} \\
 2
 \end{array}$$

$$= \int_3^4 x+3 + \frac{2}{x-2} dx \quad (1)$$

$$= \left[\frac{x^2}{2} + 3x + 2 \ln|x-2| \right]_3^4 \quad (1)$$

$$= \frac{13}{2} + 2 \ln 2 \quad (1)$$

$$4/ (a) \quad 3x+7 = A(x+2)(x+3) + B(x+1)(x+3) + C(x+1)(x+2)$$

$$\text{let } x = -2$$

$$\therefore 1 = B(-1)(1) \quad \therefore B = -1 \quad (1)$$

$$\text{let } x = -1$$

$$\therefore 4 = A(1)(2) \quad \therefore A = 2 \quad (1)$$

$$\text{let } x = -3$$

$$\therefore -2 = C(-2)(-1) \quad \therefore C = -1 \quad (1)$$

$$\therefore \frac{3x+7}{(x+1)(x+2)(x+3)} = \frac{2}{x+1} - \frac{1}{x+2} - \frac{1}{x+3}$$

$$(b) = \left[2 \ln(x+1) - \ln(x+2) - \ln(x+3) \right]_1^1 \quad (1)$$

$$= 2 \ln 2 - \ln 3 + \ln 2 - \ln 4 + \ln 3^0 \quad (1)$$

$$= 3 \ln 2 - 2 \ln 2$$

$$= \ln 2$$

$$(1)$$

$$5) \quad t = \tan \frac{x}{2} \Rightarrow dx = \frac{2dt}{1+t^2}$$

$$\begin{aligned} \text{When } x=0, \quad t=0 \\ \text{When } x=\frac{\pi}{3}, \quad t=\frac{1}{\sqrt{3}} \end{aligned} \quad (1)$$

$$= \int_0^{\frac{1}{\sqrt{3}}} \frac{1}{1 + \frac{1-t^2}{1+t^2} - \frac{2t}{1+t^2}} \cdot \frac{2}{1+t^2} dt \quad (1)$$

$$= \int_0^{\frac{1}{\sqrt{3}}} \frac{2dt}{1+t^2+1-t^2-2t}$$

$$= \int_0^{\frac{1}{\sqrt{3}}} \frac{2}{2(1-t)} dt$$

$$= \left[-\ln |1-t| \right]_0^{\frac{1}{\sqrt{3}}} \quad (1)$$

$$= -\ln \left(1 - \frac{1}{\sqrt{3}} \right) + \ln 1$$

$$= -\ln \left(\frac{\sqrt{3}-1}{\sqrt{3}} \right) \quad (1)$$

$$= \ln \left| \frac{\sqrt{3}}{\sqrt{3}-1} \right|$$

$$\cancel{6}/(a) \sin a \cos b + \cos a \sin b + \sin a \cos b - \cos a \sin b = 2 \sin a \cos b \quad (1)$$

(b) From (a)

$$\sin A \cos B = \frac{1}{2} [\sin(A+B) + \sin(A-B)]$$

$$\therefore \int_0^{\frac{\pi}{4}} \sin 5x \cos 3x dx = \frac{1}{2} \int \sin 8x + \sin 2x dx \quad (1)$$

$$\text{as } A = 5x, B = 3x$$

$$= \frac{1}{2} \left[-\frac{1}{8} \cos 8x - \frac{1}{2} \cos 2x \right]_0^{\frac{\pi}{4}}$$

$$= \frac{1}{2} \left[-\frac{1}{8} \cos 2\pi + \frac{1}{8} - \frac{1}{2} \cos \frac{\pi}{2} + \frac{1}{2} \right]$$

$$= \frac{1}{2} \left[-\frac{1}{8} - 0 + \frac{1}{8} + \frac{1}{2} \right]$$

$$= \frac{1}{4}$$

(1)

$$7/a) \quad u = x^n$$

$$\frac{du}{dx} = nx^{n-1}$$

$$\frac{dv}{dx} = e^{-x}$$

$$v = -e^{-x} \quad (1)$$

$$I_n = \left[-e^{-x} \cdot x^n \right]_0^1 + \int_0^1 nx^{n-1} \cdot e^{-x} dx \quad (1)$$

$$= \left[-e^{-1} \cdot 1 - 0 \right] + n I_{n-1}$$

$$= -\frac{1}{e} + n I_{n-1} \quad (1)$$

$$(b) \quad I_3 = -\frac{1}{e} + 3 I_2$$

$$I_2 = -\frac{1}{e} + 2 I_1$$

$$I_1 = -\frac{1}{e} + I_0 \quad (1)$$

$$I_0 = \int_0^1 e^{-x} dx$$

$$= \left[-e^{-x} \right]_0^1$$

$$= 1 - \frac{1}{e}$$

(1)

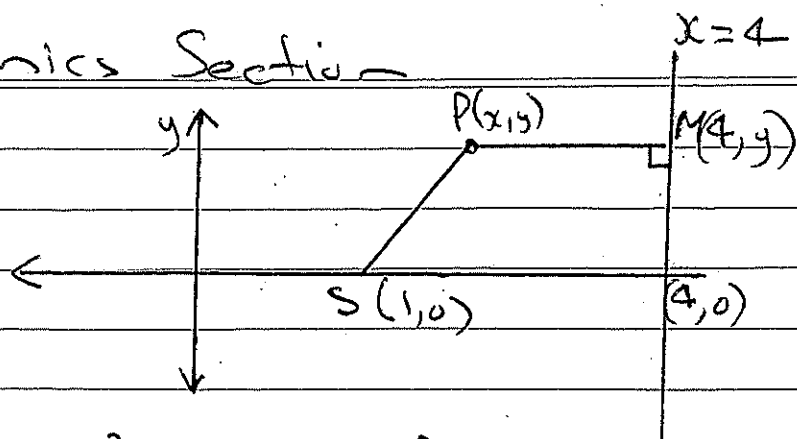
$$\therefore I_1 = 1 - \frac{2}{e}$$

$$I_2 = 2 - \frac{3}{e}$$

$$\text{hence } I_3 = 6 - \frac{16}{e} \quad (1)$$

Conics Section

1)



$$(PS)^2 = \frac{1}{4} (PM)^2$$

$$(x-1)^2 + y^2 = \frac{1}{4} (x-4)^2$$

(1)

$$x^2 - 2x + 1 + y^2 = \frac{1}{4} (x^2 - 8x + 16)$$

$$4x^2 - 8x + 4 + 4y^2 = x^2 - 8x + 16$$

$$3x^2 + 4y^2 = 12$$

(1)

$$\frac{x^2}{4} + \frac{y^2}{3} = 1$$

2) $xy = 12$

$$c^2 = 12$$

$$c = \sqrt{12}$$

$$e = \sqrt{2}$$

$$\text{Foci } \pm (\sqrt{2c}, \sqrt{2c}) = \pm (\sqrt{24}, \sqrt{24}) \quad (1)$$

$$= \pm (2\sqrt{6}, 2\sqrt{6})$$

$$\text{Directrices } x + y = \pm \sqrt{2c}$$

$$= \pm \sqrt{24}$$

$$= \pm 2\sqrt{6}$$

(1)

$$3/(a) \quad \frac{x^2}{a^2} - \frac{(px+q)^2}{b^2} = 1$$

$$bx^2 - a^2(p^2x^2 + 2pqx + q^2) = a^2b^2 \quad (1)$$

$$(b^2 - a^2p^2)x^2 - 2a^2pqx - (a^2q^2 + a^2b^2) = 0$$

tangent $\therefore \Delta = 0$

$$4a^4p^2q^2 - 4(b^2 - a^2p^2)x - a^2(q^2 + b^2) = 0 \quad (1)$$

Divide by $4a^2$

$$a^2p^2q^2 + (b^2 - a^2p^2)(q^2 + b^2) = 0$$

$$a^2p^2q^2 + b^2q^2 + b^4 - a^2p^2q^2 - a^2p^2b^2 = 0$$

$$b^2(q^2 + b^2 - a^2p^2) = 0$$

$$q^2 + b^2 - a^2p^2 = 0 \quad \text{as } b \neq 0$$

$$\therefore q^2 = a^2p^2 - b^2 \quad (1)$$

(b) $a^2 = 4 \quad b^2 = 15$

tangent through $(1, 3)$

$$\therefore 3 = p + q \Rightarrow p = 3 - q \quad (1)$$

Using (a) $q^2 = 4p^2 - 15$

$$q^2 = 4(3 - q)^2 - 15$$

$$q^2 = 4(9 - 6q + q^2) - 15$$

$$0 = 3q^2 - 24q + 21 \quad (1)$$

$$0 = q^2 - 8q + 7$$

$$0 = (q - 7)(q - 1)$$

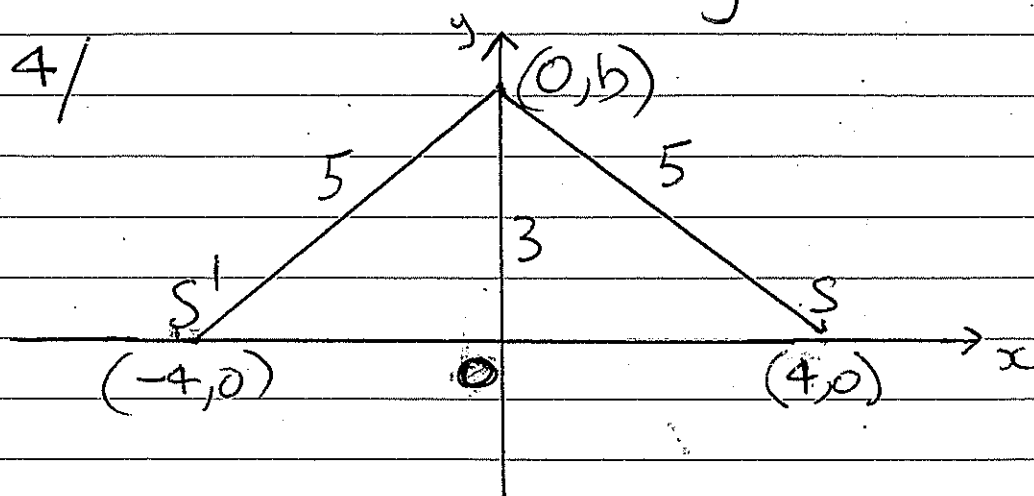
$$\therefore q = 7 \quad \text{and} \quad q = 1$$

$$p = -4$$

$$p = 2 \quad (1)$$

∴ targets are $y = -4x + 7$

and
 $y = 2x + 1$



When P is at $(0, b)$, $b = 3$ (by Pythag)

When P is at $(a, 0)$, $a = 5$

as $\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1$

E is $\frac{x^2}{25} + \frac{y^2}{9} = 1$

(ii) $x = 5 \cos \theta$ $\cos \theta = \frac{x}{5}$

$y = 3 \sin \theta$ $\sin \theta = \frac{y}{3}$

∴ $\frac{x^2}{25} + \frac{y^2}{9} = \cos^2 \theta + \sin^2 \theta$
 $= 1$

(1)

$$(c) \quad \frac{dx}{d\theta} = -5\sin\theta \quad \frac{dy}{d\theta} = 3\cos\theta$$

$$\frac{dy}{dx} = \frac{3\cos\theta}{-5\sin\theta} \quad \text{When } \theta = \frac{\pi}{6}$$

$$m_T = \frac{3 \times \frac{\sqrt{3}}{2}}{-5 \times \frac{1}{2}}$$

$$x = \frac{5\sqrt{3}}{2}$$

$$y = \frac{3}{2}$$

(1)

$$= \frac{3\sqrt{3}}{2} \times \frac{2}{-5}$$

$$= -\frac{3\sqrt{3}}{5}$$

$$\therefore m_n = \frac{5}{3\sqrt{3}}$$

(1)

$$y - \frac{3}{2} = \frac{5}{3\sqrt{3}} \left(x - \frac{5\sqrt{3}}{2} \right)$$

$$2y - 3 = \frac{10}{3\sqrt{3}} \left(x - \frac{5\sqrt{3}}{2} \right)$$

(1)

$$\begin{aligned} 6\sqrt{3}y - 9\sqrt{3} &= 10x - 25\sqrt{3} \\ &= 10x - 6\sqrt{3}y - 16\sqrt{3} \\ &= 5x - 3\sqrt{3}y - 8\sqrt{3} \end{aligned}$$

$$(d) \quad e = \frac{4}{5}$$

(1)

$$\text{directrices are } x = \pm \frac{25}{4}$$

(1)